

Job Entry Subsystems

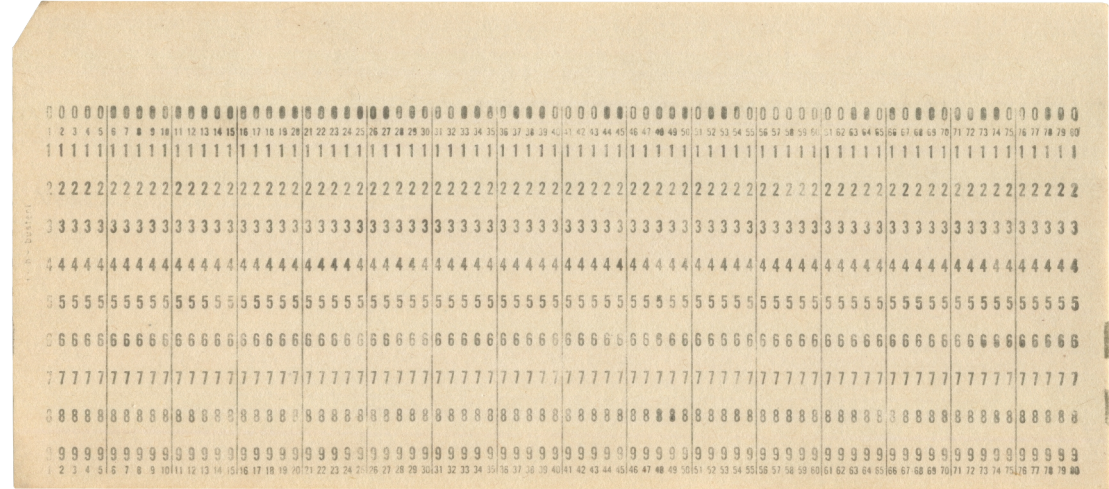




WHAT IS A JOB ENTRY SUBSYSTEM?

The term "job entry subsystem" is (as far as I know) unique to the IBM mainframe world.

According to the IBM documentation, its purpose is to "receive jobs into the operating system, schedule jobs for processing by MVS, and control job output processing."



In historical context, "receive jobs into the operating system" originally meant *read punched cards containing JCL statements and other records*.

When punched cards became obsolete and job submission through terminals became the norm, JES2 added the capability to read card images from remote terminals. That functionality is handled by an *internal reader*. It takes in 80-byte card-image records and presents them to the Brain-in-a-Jar as if they were punched cards.

WHAT IS A JOB ENTRY SUBSYSTEM?



"Schedule jobs for processing by MVS" originally meant *identify the individual jobs encoded on the punched cards and schedule them for processing.*

You've experienced writing JCL streams that contain more than one JOB statement. Imagine a stack of 2,000 punched cards; it could easily contain several jobs with different priorities and operational characteristics.

"Control job output processing" originally referred only to job output destined for printers. When printers became (mostly) obsolete, that output ended up on DASD where users could look at it *via* a terminal. But it was never *all* the output a job might produce. It was just potentially-printable output.

Finally, the word subsystem tells us JES operates as a subsystem and not as a regular job. As you know, it's the *primary* subsystem on z/OS.

WHAT IS A SPOOLER?

Peripheral devices transfer data much slower than computers can process the data. As programs rapidly generate output meant for printing, the output accumulates waiting for the much slower printers to consume it.

In the early years of mainframes, people came up with various ways to store that data temporarily while their printers caught up with it. Some used a smaller, secondary computer to handle it. But the main technology to stage printer-bound files was the tape drive.

On *spools* of tape.

The type of software that copied output from programs onto tape and read it back again to feed it to printers was...wait for it...a *spooler*.

Today the same sort of output is stored on solid state or optical or other high-tech devices. There isn't a spool in sight. But that function of JES is still called spooling, and the data sets where the data goes are called spooler data sets. Today, they're VSAM Linear Data Sets.



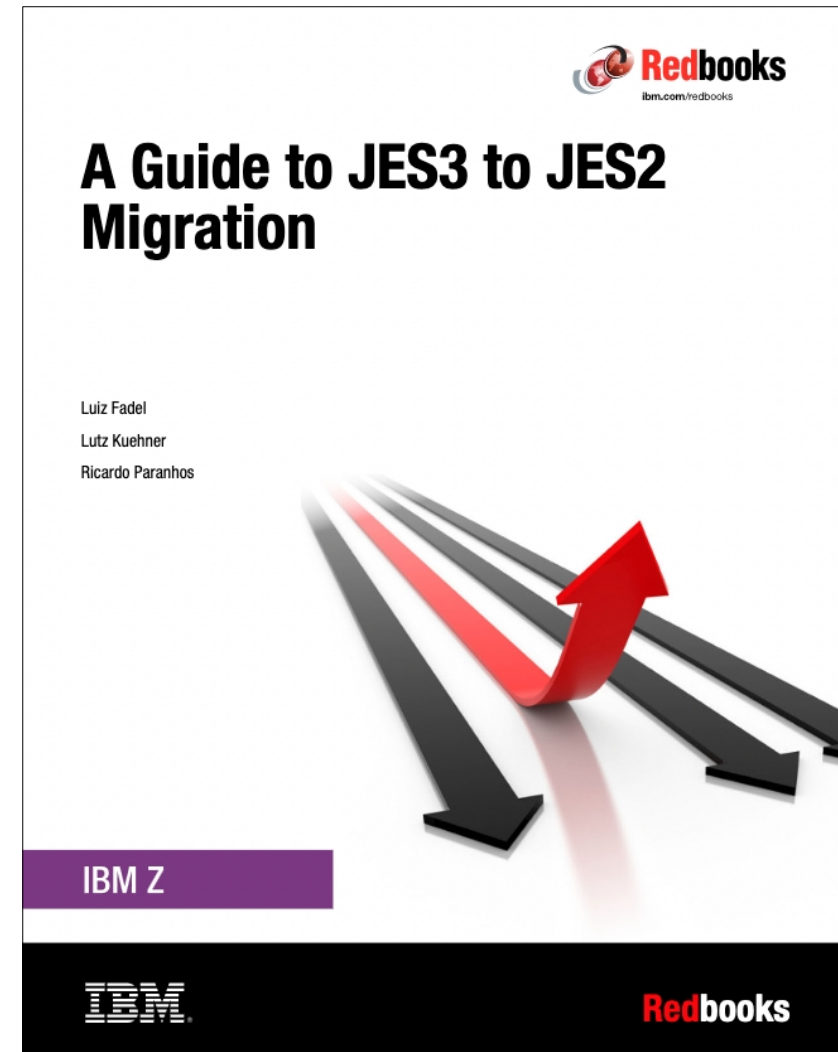
JES VERSIONS

The names of IBM's Job Entry Subsystem products may be a little misleading. One would assume JES2 is "version 2" of JES, and JES3 is "version 3."

But the numbers 2 and 3 are not sequential version numbers. They are two different products, both of which perform the function of managing "job entry" on z/OS systems.

Many components and subsystems of z/OS began life as solutions written "in the field" to support a computing need of a particular IBM customer. In more than one case, multiple solutions to the same problem were developed. Quite often, two or more survived, as different IBM customers preferred different products.

Job entry / job scheduling / output spooling is one example. IBM's SPOOL System predates the System/360. Later developments – the Houston Automatic Spooling Priority (HASP), which evolved into JES2, and the Attached Support Processor (ASP), which evolved into JES3 – took spooling technology forward. IBM later coined the acronym SPOOL – Simultaneous Peripheral Operations On-Line – which many people believe is a backronym. So, JES2 and JES3 are spoolers.



JES3 DEPRECATED AS OF Z/OS 2.5



The time came when IBM dropped support for JES3. Supporting two products that do the same job doesn't seem like a sound financial strategy.

But what about that promise of backward compatibility forever?

The promise applies to the main line of product development. z/OS today, running on modern Z hardware, doesn't run as a single, isolated instance anymore (well, it *can* be configured that way, but it in that case it runs as a virtualized instance in a local Sysplex, without really making use of Sysplex capabilities.) JES2 is optimized for that.

JES3 reached end-of-life as of z/OS 2.5. Now we're well into version 3. JES3 isn't just deprecated. It's gone to meet its maker.

JES3 TO JES2 MIGRATION CHALLENGES

So, why not "just" migrate to JES2?

Reason 1:

Over the years the two products were maintained and developed independently of each other. Useful features were added to JES3 that were not supported in JES2. Customers chose JES3 over JES2 because they considered it the superior product, and not because they were a "small" installation.

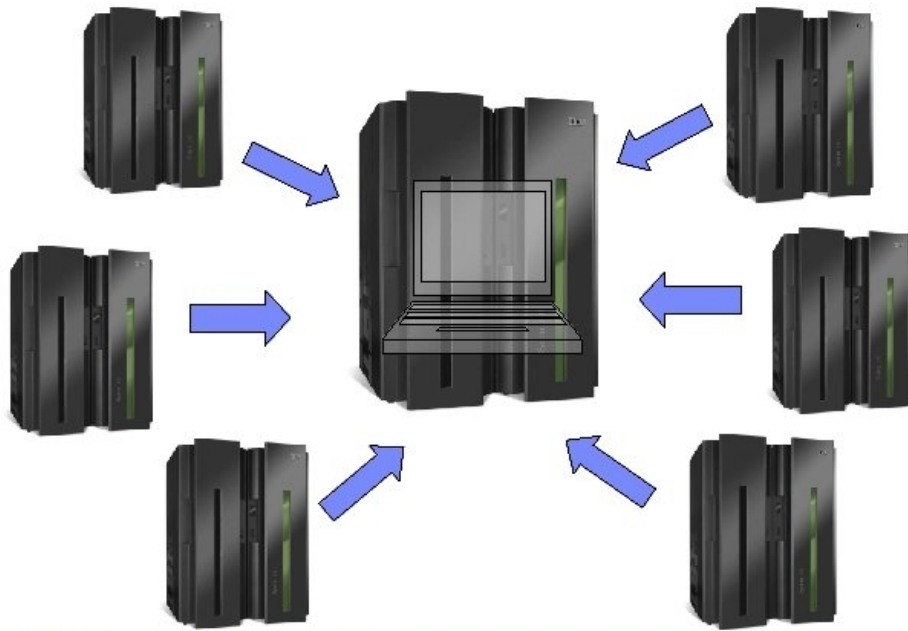
The functionality of z/OS itself continued to be enhanced, and it has built-in alternatives to these JES3 features. Yet, customers who have built tooling and operational procedures around JES3 can't easily change everything to eliminate dependency on JES3. A migration of that scope is costly and time-consuming.

Customers who are happy with JES3 would rather *not* migrate to JES2. Yet, if they don't do something about the JES3 deprecation, they'll be stuck at a version of z/OS that just gets older and older with each passing year. Eventually the bill will arrive, and it will never be smaller than it is today.



JES3 TO JES2 MIGRATION CHALLENGES

Sysplex Goal is a **Single System Image**



Reason 2:

Because IBM has chosen to focus new development on JES2, JES3 has some technical limitations that may create operational issues and increased costs for organizations that are continuing to grow their mainframe operations.

The main one is that JES2 is designed to operate in a Parallel Sysplex environment while JES3 is not. JES2 can be configured as a Multi-Access Spool (MAS), also called a JESplex. It supports up to 32 z/OS instances, with JES2 MAS members able to join and leave the MAS dynamically. This spreads the workload over the whole Sysplex and enables some degree of specialization for some of the JESplex members. Meanwhile, JES3 continues to operate out of a single address space, even as the Sysplex grows.

If the installation is growing, it may be developing from a "small" mainframe shop to a "large" one, and literally out-growing JES3. The operational cost of sticking with JES3 will only increase with time.

Of course, many mainframe installations aren't growing. They're only sustaining certain legacy applications that are difficult or impossible to replace. If they can keep JES3 going, they won't have to invest in a costly migration.

JES3 TO JES2 MIGRATION CHALLENGES

Reason 3:

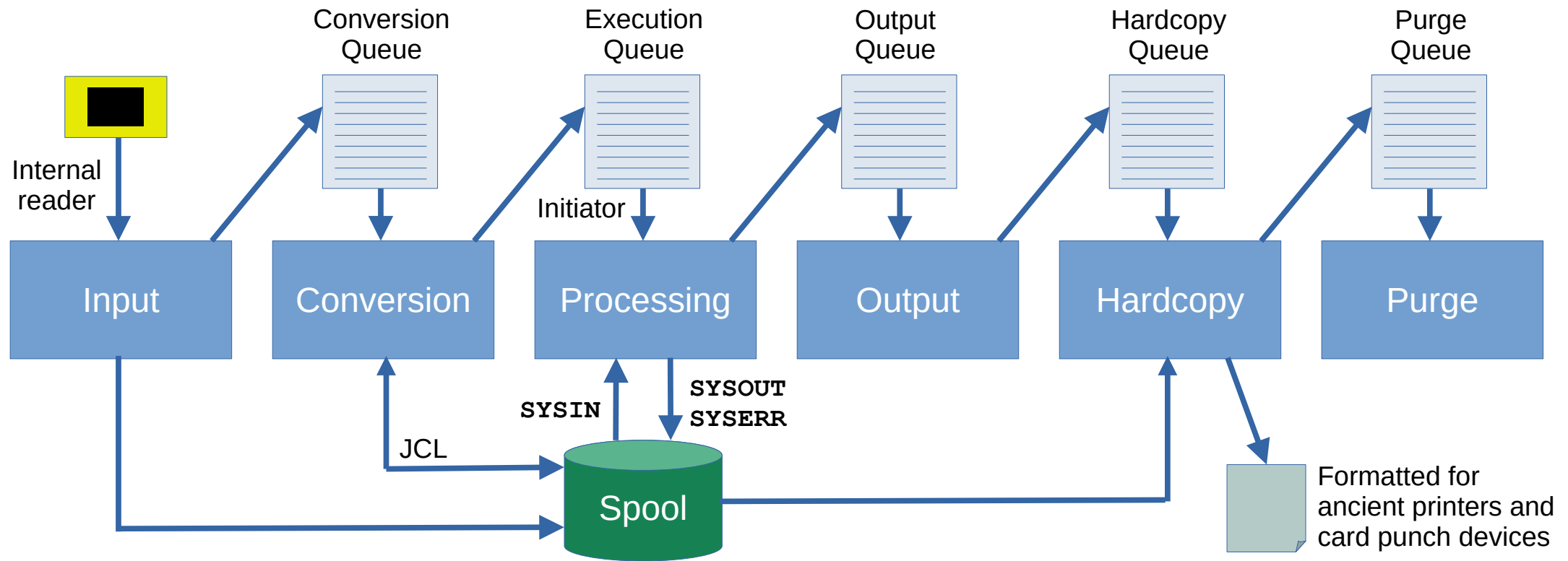
A few customers have made deep investments in customized, in-house solutions based on JES3 functionality.

An IBM Community article by Edward Jaffe, posted in 2026, describes a customer in just that situation. To facilitate their upgrade to z/OS 3.1, they purchased a third-party product that emulates JES3, JES3Plus from Phoenix Software International. It is an alternative to a difficult migration that may relieve the pressure on JES3 customers to migrate to JES2 quickly. JES3Plus is actively developed and is acquiring functionality of JES2 as well as improved Sysplex support.



JOB LIFECYCLE ON Z/OS

A job goes through several phases of processing. The z/OS base program handles execution of the code and memory management and so forth while JES2 is responsible for preparing the job for execution and managing the spooler output from the job.



JES CUSTOMIZATION

Both JES products are highly customizable. There are numerous configuration settings, user exits, macros, Job Entry Control Language (JECL) statements you can include in JCL streams, and it's even possible to modify and reassemble (build) portions of the code.

You can influence the way JCL is converted, how jobs are prioritized and scheduled, and how SYSOUT and SYSERR data sets are disposed.

In the era when the main work of a computer was batch processing and nearly all output was printed on paper or punched on cards, this level of control was useful in large organizations with complex operations.

If you peruse the JES2 or JES3 documentation, you'll see most of the JECL statements are concerned with whom to notify when the job completes, or with telling the operator where to send the printout. But the world doesn't work that way anymore. Frankly, the last time I saw anyone physically print SYSOUT data sets was in the 1980s. While JES remains a critical component of a z/OS system, the value of delving into JES customization is questionable.



WHAT YOU KNOW AND WHAT YOU CAN DO

- You know the stages in the lifecycle of a job on z/OS, and the role JES plays in them.
- You're aware of issues JES3 customers are facing in migrating to JES2

WHAT'S NEXT

- TCP/IP on z/OS